

Z_N-Protected QCD Chronometry

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Archive reconstruction from the surviving project ledger. Historical sandbox binaries were ephemeral; see RECOVERY_PROVENANCE.md.

24. Exact Z_N orbit

Take

$$\mathcal{G} = \left[\prod_{k=0}^{N-1} SU(3)_k \right] \rtimes Z_N,$$

with one vectorlike colour triplet Ψ_k in each sector and a compact field a/f_a . The masses form the exact orbit

$$M_k(a) = M \left[1 - \epsilon \cos \left(\frac{a}{f_a} + \frac{2\pi k}{N} \right) \right].$$

All masses are positive for $0 < \epsilon < 1$. The full vacuum sums all sectors, but visible experiments probe only sector zero.

The root-of-unity identity ensures that for $m < N$,

$$\sum_{k=0}^{N-1} \cos^m \left(x + \frac{2\pi k}{N} \right)$$

is independent of x . The first nonconstant Coleman-Weinberg harmonic is therefore the N th:

$$V_{\text{CW}}(a) = \frac{N_c M^4 \epsilon^N}{8\pi^2} F_N \cos \frac{Na}{f_a} [1 + O(\epsilon)],$$

where for $N > 4$,

$$F_N = \frac{24 \cdot 2^{1-N}}{(N-1)(N-2)(N-3)(N-4)}.$$

The one-sector visible threshold derivative remains $O(\epsilon)$, while the complete-orbit vacuum mass is $O(\epsilon^N)$. This established discrete-Goldstone mechanism is adapted here to a real vectorlike QCD threshold [DasHook2020; Brzeminski2020; DiLuzio2021].

The complete orbit cancels lower vacuum harmonics while preserving a one-sector linear threshold response.

Figure 1: The complete orbit cancels lower vacuum harmonics while preserving a one-sector linear threshold response.

25. Minimal clean choice: Z_6

The smallest even $N > 4$ that supplies a symmetry-related vacuum with maximal visible slope is $N = 6$. Choose

$$x_0 = \frac{a_0}{f_a} = \frac{\pi}{2}.$$

Then $M_0 = M$ and

$$\frac{\partial}{\partial x} \ln \frac{M_0}{\chi} = \epsilon.$$

Since $F_6 = 1/160$,

$$m_a^2 = \frac{27}{320\pi^2} \frac{M^4}{f_a^2} \epsilon^6 + O(\epsilon^7).$$

The visible QCD coupling remains

$$d_g = \frac{2}{27} \frac{M_P}{f_a} \epsilon + O(\alpha_s \epsilon, \epsilon^2).$$

Compared with the unprotected mass,

$$\frac{m_{a,Z_6}^2}{m_{a,\text{naive}}^2} = \frac{9}{80} \epsilon^4.$$

This makes an astronomical-range scalar compatible with a much larger visible response. The quality conditions are severe: the exact exchange symmetry must be microscopic, and local operators such as Φ^6 must be forbidden or nonlocally suppressed. A Wilson-line or collective-pNGB completion is the preferred route [Hor2026].